

DAVID RIVA

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Summary

Full-stack software engineer specializing in AI-enabled tools and data-intensive applications. Built an award-winning agentic AI system that accelerated UI component development and developed a learning management system used by 8,000+ users, leveraging React, TypeScript, and Python. Engineered large-scale geospatial visualization systems for a university research initiative, processing 20+ TB of data with React, Flask, and MongoDB. Passionate about designing scalable, high-impact software that combines strong engineering fundamentals with practical machine-learning capabilities.

EXPERIENCE

Scale AI | *Software Engineer, Agents*

May 2026 - Present

- Designed custom Agentic AI solutions and automated eval harnesses leveraging the OpenAI Agents SDK and LangGraph, ensuring robust model performance for customer use cases
- Mentored an engineering intern through the lifecycle of a customer-facing AI project, driving a 15%+ performance increase in agent hill-climbing metrics.
- Scaled the Semantic Analysis Engine's ontology generation capacity from 20 to 800+ tables by systematically identifying and resolving 40+ critical bugs within the Python codebase.

C3 AI | *Forward Deployed Engineer, Generative AI*

Sep 2024 - Apr 2026

- Awarded 1st place at the C3 Agentic AI Hackathon 2025 for creating agentic tooling that boosted developer efficiency in UI component creation, using React, TypeScript, and an in-house LLM
- Spearheaded the development of C3 AI's training website, launching a training platform for 8,000+ learners using JavaScript, HTML, and CSS while navigating evolving requirements and tight deadlines
- Designed and built internal productivity tools to automate customer-facing processes, utilizing React, JavaScript, Next.js, and Electron.js, leading to a 80% reduction in the time to send learners feedback
- Developed and administered curriculum for 8,000+ learners in data science, machine learning, and application development using the C3 AI framework, JavaScript, and Python, following Scrum methodologies.
- Enhanced customer satisfaction by leading personalized debugging sessions in Python, TypeScript, and React, leveraging AI to accelerate issue resolution.

Colorado State University | *Full-stack Developer, University Research*

Dec 2022 - Jan 2024

- Engineered full-stack geospatial data visualizations (React, TypeScript, Python, Flask, MongoDB) for the Urban Sustain Project, providing researchers intuitive access to 20TB+ datasets
- Received the Excellence in Data Science Award at CSU's Celebrating Undergraduate Research Competition
- Collaborated in a 12-member Scrum team, managing source control with Git and developing full-stack features that helped climate scientists streamline their workflows.

Hewlett-Packard Inc. | *Machine Learning Engineer, Intern*

May 2023 - Aug 2023

- Enabled long-term market trend analysis by building three ETL data pipelines, extracting data from AWS S3 buckets, and loading it into the AWS RedShift cloud data warehouse
- Developed expertise in data transformation, filtering, and verification using SQL and Python within a relational database, ensuring accurate data for downstream analysis
- Applied machine learning models using Scikit-Learn, XGBoost, and Facebook Prophet; visualized insights with Matplotlib and Seaborn to support strategic forecasting

PROJECTS

RAG Agent Chatbot | <https://davidriva.dev/chat>

- Built an autonomous Retrieval-Augmented Generation (RAG) agentic system using LangGraph, OpenAI API, and vector embeddings for dynamic context retrieval, reasoning, and task execution; developed with TypeScript and Next.js, deployed on Vercel's edge network.
- Implemented secure, serverless orchestration endpoints featuring JWT-based access control, Redis KV for rate limiting and session memory, and scalable edge inference for real-time agent interactions.

Nature Nomads - eCommerce Platform | <https://github.com/davidjriva/Nature-Nomads>

- Built a full-stack travel booking platform using Node.js, Express.js, MongoDB, and Pug with secure Stripe payment integration. Implemented automated testing and CI/CD pipelines with Jest, SuperTest, and Github Actions to ensure reliability.

Distributed Traffic Router | <https://github.com/davidjriva/Distributed-Systems-Projects/tree/main/hw1>

- Designed and implemented a distributed network of N nodes capable of routing packets based on Dijkstra's shortest path algorithm in a simulated network topology

EDUCATION

Colorado State University

Aug 2021 - May 2024

Bachelor of Science, Computer Science (GPA: 4.0)

Fort Collins, CO

- Achievements:** Summa Cum Laude, Dean's List Recipient

TECHNICAL SKILLS

- Languages:** Java, JavaScript, TypeScript, Python, C/C++, SQL, Bash, HTML, CSS
- Frameworks:** React, Next.js, Node.js, Express.js, Mongoose, Hadoop, Apache Spark, PySpark
- Testing:** JUnit, Jacoco, Mockito, Postman
- Developer Tools:** MongoDB, Git, GitHub, TensorFlow, Keras, Databricks, Pip, Cargo, NPM, Redshift, Amazon S3 Buckets, Zenhub, VS Code, Cursor, PyCharm, IntelliJ
- Libraries:** Pandas, NumPy, Scikit-learn, Matplotlib